AI COMPETENCY PROGRAM

Your Guide to Mastering AI in 2026

Jahnel Group
Year of AI Initiative
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Why This Program Exists For You

Let's be honest: the world of AI can feel overwhelming.

Agents. RAG. Embeddings. Vector databases. Guardrails. Function calling. Multi-modal models. Thinking models. Fine-tuning. The terminology flies around in every tech article, every client conversation, every conference talk. You're expected to not only understand these concepts, but to know how they fit together and when to apply them.

That's exactly why we created this program. Not as a checkbox exercise, but as **your personal roadmap** through the AI landscape.

What You'll Gain

- **A structured mental model:** The AI Periodic Table gives you a framework to organize every AI concept you encounter. Instead of drowning in terminology, you'll see how pieces connect.
- **Revealed unknown unknowns:** You don't know what you don't know. This program systematically exposes the gaps in your knowledge so you can fill them intentionally rather than discovering them in front of a client.
- **Clear progression:** No more wondering "what should I learn next?" Each tier builds on the previous one with concrete skills and assessments.
- **Credibility in conversations:** When a client asks about AI capabilities, you'll speak from genuine understanding, not memorized buzzwords.
- **Career differentiation:** AI proficiency is becoming table stakes. This program documents your competency in a way that matters for your professional growth.
- **Practical application:** Every element in this program connects to real work. You'll build things, not just study theory.

Why This Matters to Jahnel Group

We've declared 2026 the "Year of AI" at Jahnel Group. That's not marketing fluff. It's a strategic commitment based on a simple truth: our clients are asking for AI capabilities, and we need to deliver with confidence.

Our Strategic Goals

- **Measurable capability:** When we say we can deliver AI solutions, we need to back that up with quantifiable evidence. This program gives us that.
- **Client confidence:** Enterprise clients need to trust that we understand what we're building. This program helps provide that assurance.
- **Staffing intelligence:** We need to know who can work on AI projects. Program levels make staffing decisions clearer and more objective.

- **Competitive positioning:** "80% of our developers are AI-certified at Practitioner level or above" is a differentiator in proposals.
- **Organizational learning:** As you progress through the program, you'll contribute to our collective knowledge base, making everyone better.

This isn't about checking boxes. It's about building genuine competency that lets us take on ambitious projects and deliver exceptional results.

The AI Periodic Table: Your Mental Model

Just as chemistry's periodic table organizes elements into families with predictable properties, the AI Periodic Table organizes AI concepts into families and complexity levels. Once you understand this structure, you can decode any AI architecture, any product demo, any technical pitch.

Credit: The AI Periodic Table concept was created by **IBM Technology**. We have adapted and extended their framework for our program.

Source: *"AI Periodic Table Explained: Mapping LLMs, RAG & AI Agent Frameworks"*
https://www.youtube.com/watch?v=ESBMgZHzfG0

The Structure

Rows represent complexity levels, from atomic primitives to cutting-edge emerging technologies:

1. **Row 1 - Primitives:** The atomic building blocks that can't be broken down further
2. **Row 2 - Compositions:** Combinations of primitives that form practical tools
3. **Row 3 - Deployment:** Production-ready systems and practices
4. **Row 4 - Emerging:** Cutting-edge developments still rapidly evolving

Groups (columns) represent functional families - concepts that serve similar purposes at different complexity levels.

The Table

	G1 Reactive	**G2** Retrieval	**G3** Orchestration	**G4** Validation	**G5** Models
Row 1 Primitives	**Pr** Prompts	**Em** Embeddings	**Cw** Context	**Ev** Evaluation	**Lg** LLM
Row 2 Compositions	**Fc** Function Call	**Vx** Vector DB	**Rg** RAG	**Gr** Guardrails	**Mm** Multi-modal
Row 3 Deployment	**Ag** Agents	**Ft** Fine-tuning	**Fw** Frameworks	**Rt** Red Team	**Sm** Small Models
Row 4 Emerging	**Ma** Multi-agent	**Sy** Synthetic	**Mc** MCP	**In** Interpret.	**Th** Thinking

Element Reference

Group 1: Reactive Family

These elements are "reactive" because small changes produce dramatically different outputs. This is the action family - from giving instructions to autonomous operation.

Element	Description
Pr Prompts	Instructions given to an AI model. The fundamental interface between human intent and AI capability. One word change can completely transform output.
Fc Function Call	When an LLM invokes external tools or APIs to take action. The model decides which function to call and with what parameters. Bridges AI reasoning to real-world systems.
Ag Agents	Autonomous AI systems that use think-act-observe loops. Given a goal, they plan steps, execute actions (often via function calls), observe results, and iterate until complete.
Ma Multi-agent	Multiple AI agents working together - debating, collaborating, specializing. One agent researches, another writes, another critiques. Emergent capability from coordination.

Group 2: Retrieval Family

These elements handle memory and knowledge - how AI systems store, find, and adapt information. Three time scales of memory: runtime, persistent, and baked-in.

Element	Description
Em Embeddings	Numerical representations of meaning. Text becomes vectors where similar meanings have similar numbers. The foundation of semantic search and memory.
Vx Vector DB	Databases optimized for storing and querying embeddings. Store millions of vectors, find the most semantically similar ones in milliseconds.
Ft Fine-tuning	Adapting a base model by training on specific data. Bakes knowledge directly into the model's weights. Domain expertise becomes part of the model itself.
Sy Synthetic	Using AI to generate training data for AI. When real examples are scarce or expensive, synthetic data fills the gap. Increasingly important as data demands grow.

Group 3: Orchestration Family

These elements coordinate multiple components into working systems. You can't orchestrate one thing - this family only exists through combination.

Element	Description
Cw Context	Context windows - the limited space an LLM can "see" at once. Managing what goes into context is fundamental orchestration. Every token costs money and attention.
Rg RAG	Retrieval-Augmented Generation. Question comes in, relevant context is retrieved (via embeddings/vector DB), prompt is augmented with that context, LLM generates grounded answer.
Fw Frameworks	Platforms like LangChain, LlamaIndex, and others that provide the plumbing to connect AI components. Handles the complexity of building and deploying AI systems.
Mc MCP	Model Context Protocol and similar standards. Emerging protocols for how AI systems connect to tools and data sources. The USB standard of AI integration.

Group 4: Validation Family

These elements ensure AI systems work correctly and safely. From measuring quality to preventing misuse to understanding why models behave as they do.

Element	Description
Ev Evaluation	Measuring AI quality through metrics, benchmarks, and human evaluation. If you can't measure it, you can't improve it. The foundation of all quality work.
Gr Guardrails	Runtime safety filters, schema validation, content filtering. Making sure AI doesn't say things it shouldn't or output malformed garbage. Production necessity.
Rt Red Team	Adversarial testing - actively trying to break the AI. Jailbreaks, prompt injection, data exfiltration attempts. Finding vulnerabilities before attackers do.
In Interpret.	Interpretability - understanding why models do what they do. Peering inside the black box, finding neurons responsible for specific behaviors. Frontier safety research.

Group 5: Models Family

The noble gases of AI - stable foundational capabilities that everything else reacts around. The raw intelligence that powers the entire table.

Element	Description
Lg LLM	Large Language Models - GPT-4, Claude, Gemini, and others. The core reasoning engines trained on vast text. The primitive capability everything else builds on.

Mm Multi-moda l	Models that process multiple input types - text, images, audio, video. See a chart and explain it. Hear a question and answer it. Unified intelligence across modalities.
Sm Small Models	Distilled, specialized models. Fast, cheap, run on phones. When you don't need frontier capability, small models deliver 90% of value at 10% of cost.
Th Thinking	Models that reason before answering. Chain-of-thought built into architecture. Spend compute time thinking, not just generating. The smartest models today.

Certification Tiers Overview

The certification program has three progressive tiers. Each builds on the previous, with increasing depth of understanding and practical capability.

	Foundation	Practitioner	Expert
Core Question	Can you understand and use AI effectively?	Can you build and deploy AI features?	Can you architect AI systems and lead others?
Primary Rows	Row 1 (Primitives) + Row 2 concepts	Row 2 (Compositions) + Row 3 intro	Row 3 (Deployment) + Row 4 (Emerging)
Timeline	4-6 weeks	3-4 months	6-12 months
Portfolio	3 documented AI use cases	1 production feature shipped	1 architecture led + mentorship
Assessment	Written exam + portfolio review	Technical demo + code review	Architecture presentation + peer review

Think of it as learning to drive: Foundation is understanding the rules and controls. Practitioner is being able to drive anywhere safely. Expert is teaching others to drive and designing better roads.

Tier 1: Foundation

> ***"I can have an intelligent conversation about AI and use it effectively."***

What This Level Means

Foundation certification demonstrates that you understand the core concepts of modern AI, can use AI tools effectively in your daily work, and can engage meaningfully in technical conversations about AI with colleagues and clients. You know the vocabulary, understand the capabilities and limitations, and can make informed decisions about when and how to apply AI.

Elements to Master

Element	Concept	Competency	Assessment
Pr	Prompts	Write effective prompts with clear instructions, context, examples, and constraints. Understand prompt patterns.	Submit 5 prompts demonstrating structured technique with iteration notes.
Lg	LLMs	Understand what LLMs are, how they work at a high level, their capabilities, limitations, and hallucination risks.	Written assessment on model characteristics and appropriate use cases.
Em	Embeddings	Conceptual understanding of how semantic similarity works and why it matters for AI applications.	Explain embeddings to a non-technical stakeholder in writing or recorded video.
Gr	Guardrails	Understand AI safety, bias, ethical considerations, and why guardrails matter in production systems.	Case study analysis: identify risks and propose mitigations for a given AI application.
Rg	RAG	Understand the RAG pattern conceptually: retrieval, augmentation, generation. Know when and why to use it.	Diagram a RAG architecture and explain the data flow for a given use case.
Ev	Evaluation	Understand how AI quality is measured. Know common metrics, benchmarks, and	Propose an evaluation strategy for a given AI feature.

		the importance of human evaluation.	

Portfolio Requirements

Document 3 real work tasks where you used AI effectively. For each, include:

- The task or problem you were solving
- Your prompting approach and iterations
- The outcome and how AI contributed
- What you learned or would do differently

Timeline & Assessment

- **Expected duration:** 4-6 weeks of self-paced learning
- **Written assessment:** Pass with 80% or higher
- **Portfolio review:** Reviewed by a certified Practitioner or Expert

Tier 2: Practitioner

> ***"I can build and deploy AI-powered features in production systems."***

Prerequisites

Foundation certification required before beginning Practitioner track.

What This Level Means

Practitioner certification demonstrates that you can independently build AI-powered features and deploy them to production. You understand the technical implementation details, can make architecture decisions for standard AI patterns, and can troubleshoot issues. You're ready to take on AI work on client projects with appropriate guidance.

Elements to Master

Element	Concept	Competency	Assessment
Fc	Function Calling	Implement tool use and API integration with LLMs. Design function schemas, handle responses, manage errors.	Build a working demo that uses function calling to interact with external APIs.
Vx	Vector Databases	Set up and query vector databases. Understand indexing strategies, similarity metrics, and performance tradeoffs.	Implement semantic search on a real dataset with measurable retrieval quality.
Rg	RAG (Advanced)	Build complete RAG pipelines. Implement chunking strategies, retrieval optimization, and context management.	Deploy a RAG system against documentation with demonstrated accuracy.
Mm	Multi-modal	Work with vision and audio models. Understand input processing, use cases, and limitations.	Build a feature using image or audio input with appropriate error handling.
Ag	Agents	Implement basic agentic loops. Understand planning,	Create an agent that completes a multi-step task with observable reasoning.

		tool selection, observation, and termination conditions.	
Fw	Frameworks	Proficiency in at least one major AI framework. Understand abstractions, patterns, and when to use them.	Deliver a project using LangChain, LlamaIndex, or equivalent framework.
Sm	Small Models	Know when and how to use smaller models. Understand cost/latency/quality tradeoffs.	Demonstrate model selection analysis with cost and performance comparison.
Cw	Context Windows	Manage context effectively. Understand token limits, context stuffing, and optimization strategies.	Implement context management for a long-document use case.

Portfolio Requirements

Ship at least one AI-powered feature to production (internal or client). Document:

- Architecture decisions using periodic table elements
- Technical implementation details
- Challenges encountered and how you solved them
- Measurable outcomes (accuracy, latency, cost, user feedback)

Timeline & Assessment

- **Expected duration:** 3-4 months
- **Technical assessment:** Pass with 85% or higher
- **Live demo:** Demonstrate your production feature with Q&A
- **Code review:** Reviewed by a certified Expert

Tier 3: Expert

"I can architect AI systems, make strategic technology decisions, and advance organizational capability."

Prerequisites

Practitioner certification + at least 2 production AI deployments required before beginning Expert track.

What This Level Means

Expert certification demonstrates that you can architect complex AI systems, make strategic technology decisions, lead AI initiatives, and elevate others' capabilities. You understand cutting-edge developments, can evaluate emerging technologies, and can guide the organization's AI direction. You're the go-to person for hard AI problems.

Elements to Master

Element	Concept	Competency	Assessment
Ft	Fine-tuning	Prepare datasets and fine-tune models for specific domains. Understand when fine-tuning beats RAG and vice versa.	Complete a fine-tuning project with measurable improvement over base model.
Rt	Red Teaming	Adversarial testing, prompt injection defense, security assessment. Find vulnerabilities systematically.	Conduct red team exercise on existing system with documented findings.
Ma	Multi-agent	Design and implement multi-agent orchestration. Understand agent specialization, coordination, and failure modes.	Build a system with multiple specialized agents collaborating on complex tasks.
Sy	Synthetic Data	Generate and validate synthetic training data. Understand quality, diversity, and contamination risks.	Create synthetic dataset for a specific use case with quality validation.
In	Interpretability	Debug model behavior, understand attention	Root-cause analysis of a model failure with

		patterns, root-cause failures. Know the limits of explainability.	documented investigation process.
Th	Thinking Models	Architect solutions using reasoning models. Understand when to invest compute in thinking vs. generation.	Design system that optimizes reasoning vs. speed tradeoffs with documented rationale.
Mc	MCP/Protocols	Understand emerging standards for AI tool integration. Evaluate and implement protocol-based architectures.	Implement MCP or similar protocol integration with analysis of tradeoffs.

Additional Expert Competencies

Beyond specific elements, Experts demonstrate mastery in:

- **Cost modeling and optimization:** Analyze and optimize AI costs at scale
- **Architecture reviews:** Evaluate and improve AI system designs
- **Mentorship:** Guide others through Practitioner certification
- **Knowledge contribution:** Add to Jahnel Group's AI knowledge base

Portfolio Requirements

- Lead architecture on a complex AI system (internal or client)
- Present a technical deep-dive to peers
- Mentor at least one person through Practitioner certification

Timeline & Assessment

- **Expected duration:** 6-12 months
- **Technical assessment:** Pass with 90% or higher
- **Architecture presentation:** Present system design to Expert panel
- **Peer review:** Evaluation from existing Experts and leadership

Getting Started

Ready to begin? Here's your path forward:

Step 1: Self-Assessment

Review the periodic table and honestly assess where you are today. Can you explain each element in Row 1? Have you built anything with Row 2 elements? This self-assessment helps you understand your starting point and identify your unknown unknowns.

Step 2: Register for Foundation

Everyone starts with Foundation, regardless of experience. Even if you've built AI systems before, the Foundation certification ensures we have a common vocabulary and shared mental model across the organization. The assessment may surface gaps you didn't know you had.

Step 3: Build Your Portfolio as You Learn

Don't wait until you feel "ready" to start documenting your AI work. Every time you use AI effectively in your job, make a note. Every experiment, every prompt iteration, every lesson learned - these become your portfolio. The best portfolios are built over time, not crammed at the end.

Step 4: Connect with Others

Learning is faster with others. Find colleagues at your level to study together. Ask questions of those ahead of you. When you reach Practitioner and Expert levels, mentoring others is part of the certification - but informal mentoring can start anytime.

The Year of AI starts with you.

Every expert was once a beginner. Every complex system is built from simple elements. Pick up the periodic table, identify where you are, and take the next step.

Questions? Reach out!